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Intelligent Pose Estimation Framework for Fitness Form Correction

AN AI-BASED EXERCISE COACH

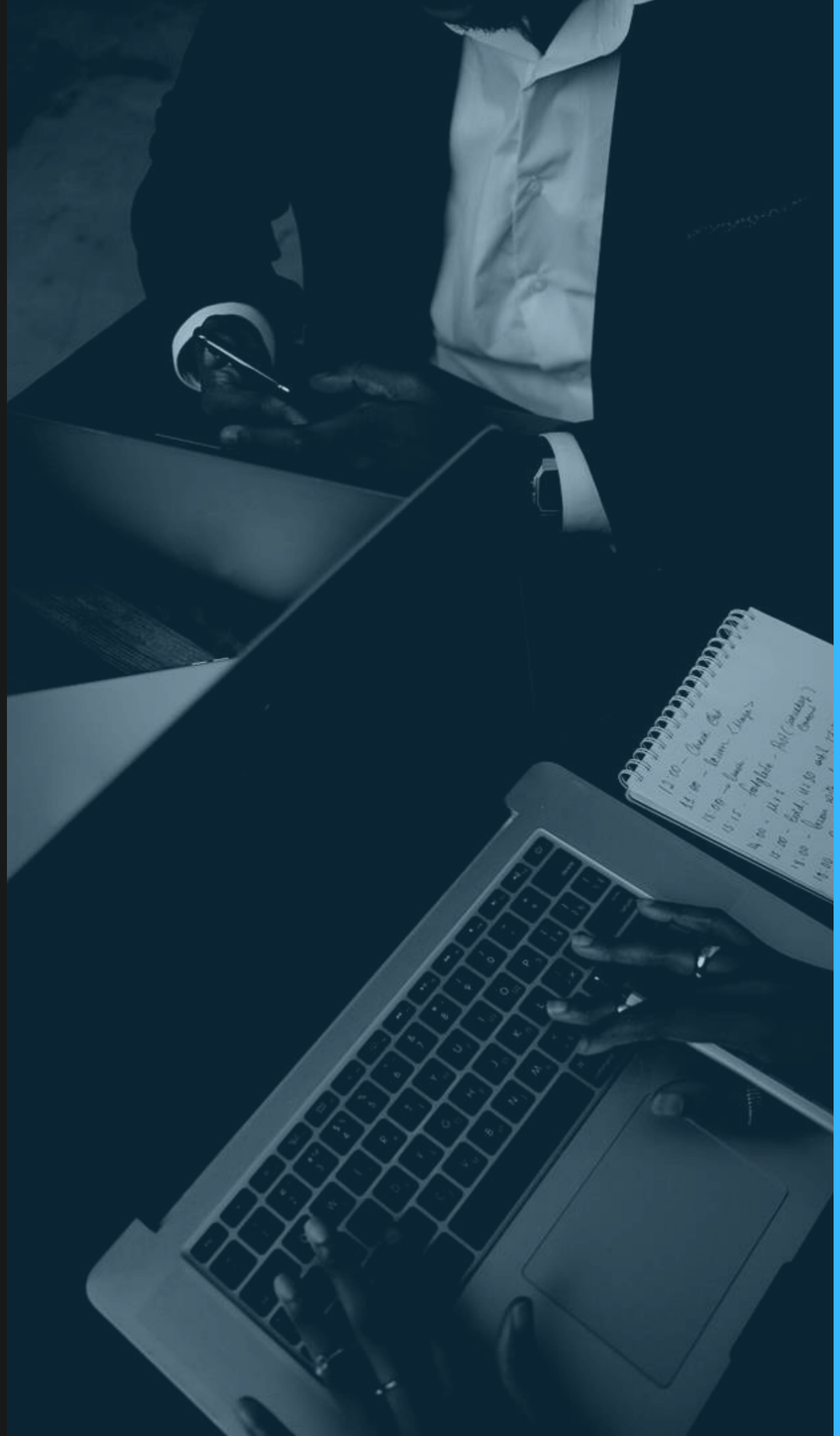
Introduction - The need for something like this

Troubles

- Injury Risk: Poor exercise form is a leading cause of sports-related injuries, leading to long-term health issues and dropout.

- Reduced Effectiveness: Incorrect execution significantly diminishes workout effectiveness, wasting effort and time.

- Cost Barrier: Professional personal trainers are expensive, while existing apps provide generic, non-adaptive feedback.



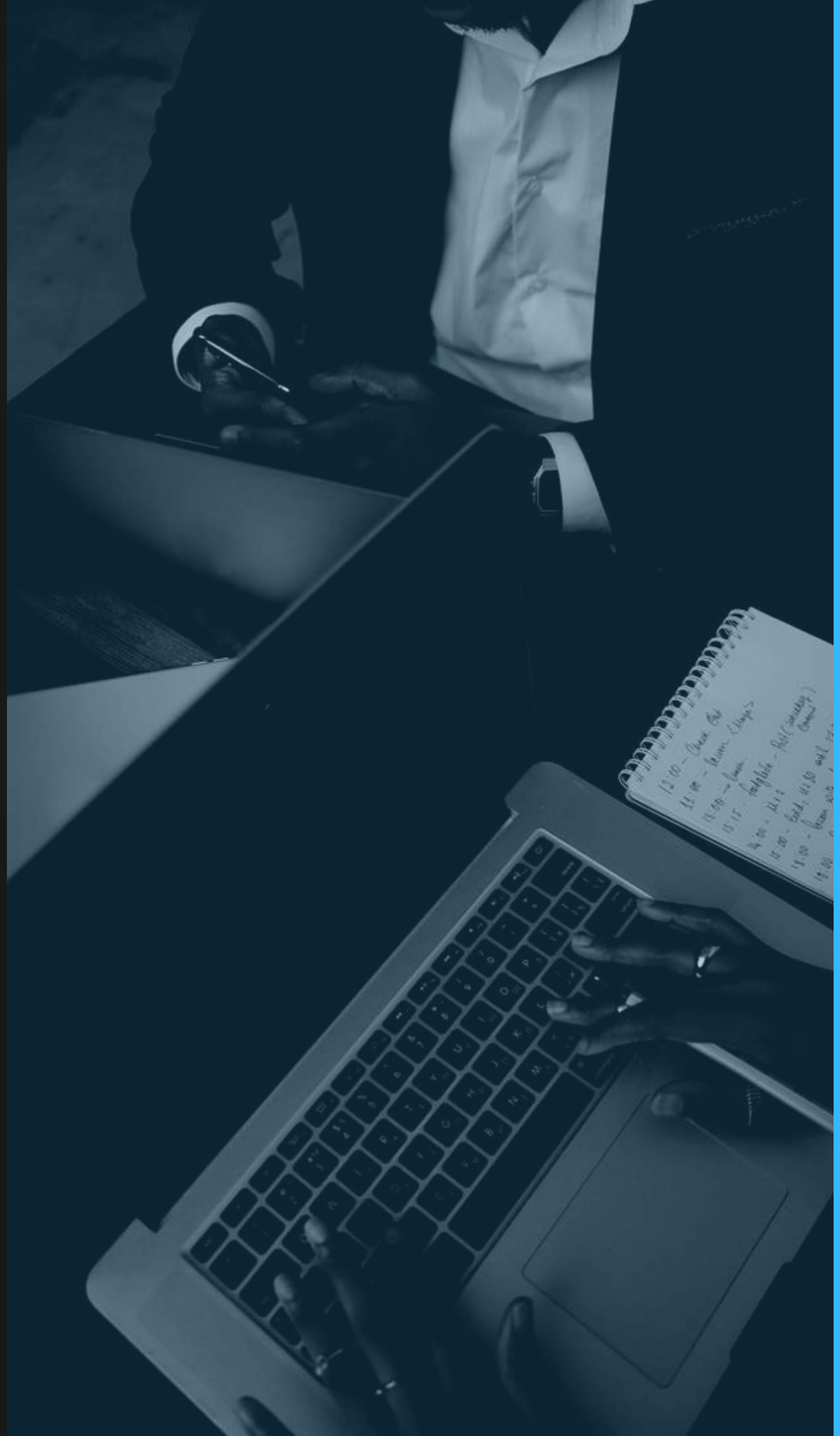
Introduction - About Pose Detection

- Real-Time Pose Detection: Uses your webcam and MediaPipe to detect 33 body landmarks (shoulders, elbows, hips, knees) in real-time with skeleton overlay.

- Live Visualization: Displays your movements with limb connections and joint points, showing exactly how the system "sees" your exercise form.

- Data Dashboard: Provides instant feedback on key joint coordinates, visibility scores, and raw landmark data for analysis.

- Accessible Design: Works with any standard webcam—no expensive hardware or subscriptions required, making AI-powered form analysis available to everyone.



Literature Review-Applied Bias Detection & Explainability

Article Title	Method Applied	Dataset Used	Conclusion
Google Research MediaPipe	Top-Down Approach	COCO, MPII	Balances Accuracy and Speed
Google AI(2021) MoveNet	Bottom-Up Approach	Google Fitness	Ultra-fast(50+fps); Good for Basic Exercises
Standforn HCI(2021) Pose Trainer	CNN + LSTM	Custom collected exercise dataset	Applies LSTM for temporal analysis

Problem Statement

Current digital fitness solutions lack real-time, personalized exercise form correction that adapts to individual users' body mechanics, leading to ineffective workouts and increased injury risk.

Objectives

Objective 1:

Develop a real-time pose estimation system using Google MediaPipe that accurately detects 33 body landmarks from standard webcam input, providing the foundation for exercise form analysis.

Objective 2:

Train and implement machine learning models (Random Forest, XGBoost) to classify exercise form quality as Good, Acceptable, or Poor based on extracted joint angles, symmetry metrics, and movement patterns, achieving ≥85% classification accuracy.

Objective 3:

Create an adaptive feedback mechanism that learns individual user's body mechanics and progress over time, providing real-time corrective cues tailored to each user's unique movement patterns and improvement trajectory.

Methodology

1. Pose Estimation Pipeline:

- Real-time body landmark detection using Google MediaPipe's BlazePose model, capturing 33 keypoints (x, y, z coordinates + visibility scores) at 30+ FPS from standard webcam input. The top-down detection approach first localizes the person, then predicts precise joint locations.

2. Feature Engineering:

Extraction of 50+ biomechanical features from raw landmarks:

•**Joint angles**: Knee, hip, elbow, shoulder, torso angles

•**Symmetry metrics**: Left-right differences for alignment assessment

•**Temporal features**: Velocity and acceleration of key joints across 30-frame windows

3. Machine Learning Classification:

- Training ensemble models for form quality assessment:

•**Random Forest** for interpretable feature importance

•**XGBoost** for optimized classification performance

•Evaluation metrics: Accuracy (target ≥85%), precision, recall, F1-score

•Cross-validation to prevent overfitting

Methodology

4. Real-Time Feedback Integration:

- Deployment of trained model in Streamlit web application:
- Frame-by-frame form classification with confidence scoring
- Visual feedback (color-coded joints, on-screen cues)
- Audio prompts for corrective guidance
- Performance monitoring (FPS tracking, latency optimization)

REFERENCES

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Thank you